MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

MG1MDCECT100 - HOME APPLIANCES AND TROUBLESHOOTING

PRACTICAL EXAMINATION

Duration: 1.5 hours **Maximum Marks: 35**

Instructions:

- Demonstrate the task
- Viva session (*Minimum 5 questions from the Practical Module*)
- Lab Report

Evaluation Criteria:

- Viva: 10 marks
- Lab Report: 10 marks
- Demonstration: 15 marks

Tasks: *(Choose any one)*

1. Solder a resistor onto a PCB. Ensure there's no excess solder, and verify the strength of the connection.
2. Use a desoldering pump to remove a component from a PCB. Evaluate the cleanliness of the removed joint and identify areas that need improvement.
3. Solder three points on a PCB and intentionally create one "dry solder" joint. Identify and troubleshoot the dry solder.
4. Solder an LED and 1K resistor to a PCB and connect proper power supply to glow the LED
5. Use a desoldering wick to remove solder from a component. Compare the effectiveness with a desoldering pump.
6. Use a multimeter to test for any unintended short circuits between two points.
7. Desolder and resolder a component. Observe and note any damage to the PCB pads.
8. In an LCD/LED TV, use a multimeter to test continuity on the fuse.
9. Demonstrate how we check power supply.
10. Demonstrate how we check capacitor in a board.
11. Test the condition of capacitor of a fan/motor .

12. Test the motor windings with a multimeter. Measure the resistance across each winding and note any discrepancies that indicate a fault.
13. Open an LED bulb and identify each internal component. Use a multimeter to check if each LED module is functional.
14. Measure the output voltages from the PSU of a home theater system.
15. Test an iron box's heating element for continuity and resistance.
16. Use a multimeter to check the motor windings in a mixer grinder. Identify possible symptoms if the windings are open or shorted.
17. Inspect the heating element and thermostat in a water heater.
18. Find the thermal fuse in a high-power appliance, test for continuity, and determine if it needs replacement.
19. Demonstrate the correct way to clean dust and residue from a PCB.
20. Apply solder and flux to a PCB connection. Observe how flux prevents dry solder joints and improves the quality of the joint.
21. Apply a protective coating to a PCB and explain how it prevents environmental damage.
22. Use a multimeter to test the earthing on an appliance.
23. Use a multimeter to test the phase voltage on an appliance/switch Box.